Macroeconomics II, Lecture VIII: Burdett-Mortensen

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Recap

- Search models offer a theory of unemployment and wage dispersion
- Today we dig deeper into the determinants of wage dispersion
- The McCall model: search frictions $+$ offer distribution $\Rightarrow$ reservation wage strategy
- It generates a sensible theory of residual wage dispersion
- But the model had no chance of matching empirical measures of wage dispersion
- And how can exogenous wage dispersion be consistent with firm wage posting in the first place (Rothschild critique/Diamond paradox)?

Today

- Today we introduce a class of models that speaks to both of these problems
  - In contrast to McCall, we will explicitly model firm behavior and study *equilibrium* wage formation

- Core idea: job ladders

- Employed workers can search while on the job and, as a consequence, they will sequentially move from worse to better jobs

- Realistic feature of most labor markets

- Generates more residual wage dispersion because ex ante identical workers differ not only because their initial offers differ, but also because some have climbed further up the job ladder than others

- Solves the Diamond paradox, as firms face a trade-off when posting wages
  - Higher wage attracts more workers
  - Higher wage reduces per-worker profit

- But first: something more on the nature of wage dispersion in the data

Agenda

1. The Abowd-Kramarz-Margolis regression
2. The Burdett-Mortensen model
   - Setup
   - Solution
   - Predictions
3. Discussion and extensions

The AKM regression

The AKM regression

- Remember Mincer regression

$$\log y_{it} = \mu_y + \beta \mathbf{X}_{it} + \epsilon_{it}$$

where, typically, $R^2 < 0.3$

- Abowd, Kramarz, and Margolis (Econometrica, 1999) were the first to use French matched employer-employee data to estimate

$$\log y_{it} = \mu_y + \beta \mathbf{X}_{it} + \theta_i + \Psi_{j(i,t)} + \epsilon_{it}$$

for individual i working for firm $j(i,t)$ at time t

- A reduced-form empirical framework for studying labor market sorting and inequality
  - Card-Heining-Kline (QJE 2013); Bloom-Guvenen-Price-Song-von Wachter (QJE 2019); Frederiksen-Hensvik-Skans (AER 2018); Eliason-Hensvik-Kramarz-Skans (JE 2023)

- θ_i is identified from variation in worker earnings within firms; $\Psi_{j(i,t)}$ is identified from changes in worker earnings when workers switch jobs

- AKM find:
  - No fixed effects: $R^2 \approx 0.3$
  - Firm fixed effects only: $R^2 \approx 0.55$
  - Person fixed effects only: $R^2 \approx 0.75$
  - Both fixed effects: $R^2 \approx 0.85$

The AKM regression

- How to interpret these findings?

- Considerable variation is due to firm and individual fixed effects
  - Some workers are persistently paid more across all their jobs than other, otherwise similar, workers
  - Some firms persistently pay more to all their employees, even though those employees are similar to workers at other firms
  - Does this mean that unobserved heterogeneity, as opposed to frictions, explain most of residual wage dispersion?

- Theory will guide our interpretation...

The Burdett-Mortensen model

Overview

- Continuous time; infinite horizon
- Measure 1 of ex-ante identical firms, measure 1 of ex-ante identical households
- One-to-many matching: each firm can employ many workers
  - Each match produces y output flow
- Separation rate σ, discount rate r, unemployment utility b
- Firms decide which wage w to post, generating the offer distribution $F(w)$
- Unemployed workers search for jobs at rate λ_u, drawing opportunities from offer distribution $F(w)$
- Employed workers search for jobs at rate λ_e, drawing opportunities from the same offer distribution $F(w)$
- Each worker's problem is to accept or reject offers
- We focus on steady state

Worker value functions

- Unemployed:

$$rU = b + \lambda_u \int_{\mathbb{W}} \max\{W(w) - U, 0\} dF(w)$$

- Employed:

$$rW(w) = w + \lambda_e \int_{\mathbb{W}} \max\{W(w') - W(w), 0\} dF(w') + \sigma(U - W(w))$$

- What are the optimal acceptance strategies?

Worker strategies

- For employed, optimal acceptance rule is to accept all wages $w' \geq w$:

$$rW(w) \quad = \quad w + \lambda_e \int_{w' \geq w} (W(w') - W(w))dF(w') + \sigma(U - W(w))$$

- $W(w)$ is increasing in w (see below) $\Rightarrow$ optimal acceptance rule of unemployed is a reservation wage w_R (as in McCall):

$$rU \quad = \quad b + \lambda_u \int_{w \geq w_R} (W(w) - U)dF(w)$$

- Characterizing worker behavior boils down to finding w_R, given F, b, λ_e, λ_u, and σ

- For now, we assume that F has bounded support; later we prove that this follows from firm optimization

Finding the reservation wage I

- Reservation wage satisfies

$$W(w_R) = U$$

- From employed worker's value function equation:

$$\begin{aligned} rW(w_R) &= w_R + \lambda_e \int_{w' \geq w_R} \big(W(w') - W(w_R)\big)\, dF(w') + \sigma(U - W(w_R)) \\ &= w_R + \lambda_e \int_{w' \geq w_R} \big(W(w') - W(w_R)\big)\, dF(w') \end{aligned}$$

- Using $rW(w_R) = rU$ and the unemployed worker's value function equation:

$$\begin{aligned} & w_R + \lambda_e \int_{w \geq w_R} (W(w) - U) dF(w) = b + \lambda_u \int_{w \geq w_R} (W(w) - U) dF(w) \\ \Leftrightarrow \quad & w_R - b = (\lambda_u - \lambda_e) \int_{w \geq w_R} (W(w) - U) dF(w) \end{aligned}$$

Finding the reservation wage II

- We have found our reservation wage equation:

$$w_R - b = (\lambda_u - \lambda_e) \int_{w \geq w_R} (W(w) - U) dF(w) \tag{1}$$

- Interpretation
  - $\lambda_u > \lambda_e \Rightarrow w_R > b$: workers find better opportunities while unemployed $\Rightarrow$ unemployed workers demand compensation for giving up the option value of search
  - $\lambda_u < \lambda_e \Rightarrow w_R < b$: workers find better opportunities while employed $\Rightarrow$ unemployed workers accept a wage below b in anticipation that they will receive better offers more quickly once employed

- To solve for w_R, we need to evaluate $\int_{w \geq w_R} (W(w) - U) dF(w)$. Two steps
  1. Compute $W'(w)$
  2. Use integration by parts

Finding the reservation wage III, computing $W'(w)$

- Rewrite the employed worker's value:

$$(r+\sigma)W(w) = w + \lambda_e \int_{w' \geq w} (W(w') - W(w))dF(w') + \sigma U$$

- Apply Leibniz's rule (using bounded support):

$$\begin{aligned}(r+\sigma)W'(w) &= 1 + \lambda_e(W(w_{max}) - W(w))\frac{\partial F(w_{max})}{\partial w} \\ &\quad -\lambda_e(W(w) - W(w))\frac{\partial F(w)}{\partial w} + \lambda_e \int_{w' \geq w} -W'(w)dF(w') \\ &= 1 - \lambda_e W'(w) \int_{w' \geq w} dF(w') \\ &= 1 - \lambda_e W'(w)(1 - F(w))\end{aligned}$$

- Hence,

$$W'(w) = \frac{1}{r + \sigma + \lambda_e(1 - F(w))} \tag{2}$$

- We have now also shown that $W(w)$ is increasing in w

Finding the reservation wage IV, integration by parts

$$
\begin{aligned}
\int_{w \geq w_R} (W(w) - U) dF(w) &= (W(w) - U)F(w)|_{w_R}^{w_{max}} - \int_{w \geq w_R} F(w) d(W(w) - U) \\
\text{(use differentiability)} &= (W(w) - U)F(w)|_{w_R}^{w_{max}} - \int_{w \geq w_R} F(w) W'(w) dw \\
\text{(add/subtract)} &= (W(w) - U)F(w)|_{w_R}^{w_{max}} - \int_{w \geq w_R} F(w) W'(w) dw \\
&\quad + (W(w) - U)|_{w_R}^{w_{max}} - (W(w) - U)|_{w_R}^{w_{max}} \\
\text{(use } W(w_R) = U,\ F(w_R) = 0,\ F(w_{max}) = 1) &= -\int_{w \geq w_R} F(w) W'(w) dw + (W(w) - W(w_R))|_{w_R}^{w_{max}} \\
\text{(evaluate second term)} &= -\int_{w \geq w_R} F(w) W'(w) dw + \int_{w \geq w_R} W'(w) dw \\
\text{(rearrange)} &= \int_{w \geq w_R} (1 - F(w)) W'(w) dw \qquad (3)
\end{aligned}
$$

Finding the reservation wage V

- Plug (2) and (3) into reservation wage equation (1):

$$\begin{aligned} w_R - b &= (\lambda_u - \lambda_e) \int_{w \geq w_R} (W(w) - U) dF(w) \\ &= (\lambda_u - \lambda_e) \int_{w \geq w_R} (1 - F(w)) W'(w) dw \\ &= (\lambda_u - \lambda_e) \int_{w \geq w_R} \frac{1 - F(w)}{r + \sigma + \lambda_e (1 - F(w))} dw \end{aligned} \tag{4}$$

- LHS increasing
- Integrand is positive: RHS decreasing
- $\Rightarrow$ (4) implicitly determines w_R given $F(w)$

Worker distribution I

- We have now characterized worker behavior, taking the offer distribution $F(w)$ as given
- Before turning to the firm problem, which determines $F(w)$, we derive the worker distribution $G(w)$, taking $F(w)$ as given
- Unemployment law of motion:

$$\dot{u} = \sigma(1 - u(t)) - \lambda_u(1 - F(w_R))u(t)$$

- Steady state

$$\begin{aligned} u &= \frac{\sigma}{\sigma + \lambda_u(1 - F(w_R))} \\ &= \frac{1}{1 + k_u(1 - F(w_R))} \end{aligned} \tag{5}$$

where $k_u = \frac{\lambda_u}{\sigma}$

Worker distribution II

- $G(w) =$ the share of workers with wage $\leq w$

- $G(w)(1-u(t)) =$ the *mass* of workers with wage $\leq w$

- Law of motion for $G(w,t)(1-u(t))$:

$$\begin{aligned}\frac{d}{dt}G(w,t)(1-u(t)) &= \lambda_u \max\{F(w)-F(w_R),0\}\,u(t) \\ &\quad -\sigma G(w,t)(1-u(t)) \\ &\quad -\lambda_e(1-F(w))G(w,t)(1-u(t))\end{aligned}$$

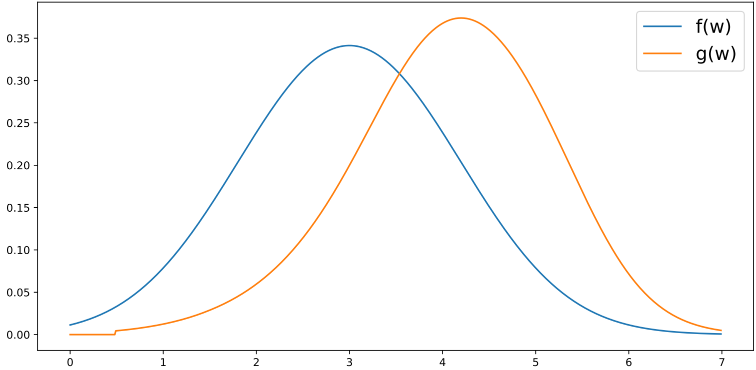
  - inflow: unemployed households that find job wage in interval $[w_R, w]$
  - outflow 1: employed households with wage $\leq w$ that separate at rate σ
  - outflow 2: employed households with wage $\leq w$ that find a job with wage $> w$

- Using steady state condition $\frac{d}{dt}G(w,t)(1-u(t)) = 0$ and Eq. (5), we solve for $G(w)$:

$$G(w) = \begin{cases} 0 & \text{if } w < w_R \\ \frac{[F(w)-F(w_R)]/(1-F(w_R))}{1+k_e(1-F(w))} & \text{if } w \geq w_R \end{cases} \tag{6}$$

where $k_e = \frac{\lambda_e}{\sigma}$

Illustration of worker distribution in steady state



- Example: assume the offer distribution $F(w)$ is normal, compute w_R from (4), and then compute $G(w)$ from (6)
- Why is pdf $g(w)$ "to the right of" pdf $f(w)$?

Firm size distribution I

- Given G and F, we can also characterize the firm size distribution
- Use a limit argument: determine how many firms offer jobs and how many workers hold jobs on the interval $[w-\epsilon, w]$, then let $\epsilon \to 0$
- Workers on interval $[w-\epsilon, w]$: $[G(w) - G(w-\epsilon)] \times (1-u)$
- Firms on interval $[w-\epsilon, w]$: $[F(w) - F(w-\epsilon)] \times 1$
- Average number of workers per firm on interval $[w-\epsilon, w]$:

$$\frac{G(w) - G(w-\epsilon)}{F(w) - F(w-\epsilon)}(1-u)$$

- Average firm size at point w:

$$l(w) = \lim_{\epsilon \to 0} \frac{G(w) - G(w-\epsilon)}{F(w) - F(w-\epsilon)}(1-u) \tag{7}$$

Firm size distribution II

- Given our solution for u and G, we can solve for l
- We have not assumed that F is continuous:
  - We don't know if $\lim_{\epsilon \to 0} F(w-\epsilon) = \lim_{\epsilon \to 0} F(w+\epsilon)$, i.e., we don't know if F has mass points
- Denote $F(w^-) = \lim_{\epsilon \to 0} F(w-\epsilon)$
- Plug in solution of G, u into firm size distribution (7), and do the algebra:

$$l(w) = \begin{cases} 0 & \text{if } w < w_R \\ \frac{k_u(1+k_e(1-F(w_R)))/(1+k_u(1-F(w_R)))}{(1+k_e(1-F(w)))(1+k_e(1-F(w^-)))} & \text{if } w \geq w_R \end{cases} \tag{8}$$

- $l(w)$ is increasing in w - why?
- $l(w)$ is continuous everywhere where F is continuous

Firm problem

- Firms maximize the expected discounted sum of profits

- For simplicity, we assume that $r \to 0 \Rightarrow$ firms only care about the steady state value of profits

- Firms post wage w to maximize

$$\pi = \max_{w} (y - w) l(w) \tag{9}$$

where $l(w)$ is the firm size distribution, taking workers' behavior (w_R) and other firms' behavior (F) as given

- Equilibrium must have $\pi(w) = \pi(w')$ for all $w, w' \in F$

- Why don't all firms set $w = w_R$ (as in the wage-posting equilibrium of the McCall model)?

Equilibrium definition

- An equilibrium is a triple $\{w_R, \pi, F\}$ such that
  1. Given F, each worker behaves optimally: w_R solves reservation wage equation (4)
  2. Given F and w_R, each firm behaves optimally: π follows from (9)
  3. $F, \mathbb{W}_F$ are such that there is no arbitrage:

$$(y-w)l(w)\begin{cases} < \pi & \text{if } w \notin \mathbb{W}_F \\ = \pi & \text{if } w \in \mathbb{W}_F \end{cases}$$

  where $\mathbb{W}_F$ is the support of F

Solution strategy

- Given the progress we have made on characterizing worker behavior and the firm size distribution, the equilibrium can be solved in five steps:
  1. Establish that F has bounded support $[\underline{w}, \bar{w}]$
  2. Establish that F is continuous
  3. Establish $\underline{w} = w_R$
  4. Solve for F
  5. Solve for w_R, π

Step 1: bounded support

- No worker accepts a wage $w < w_R \Rightarrow \pi(w < w_R) = 0$
- Also, $\pi(w > y) < 0$
- Therefore, F has bounded support $[\underline{w}, \bar{w}] \subset [w_R, y]$

Step 2: continuity

- Proof strategy: if F has a mass point at $\hat{w}$, a firm can make excess profits by offering $\hat{w}+\epsilon$

- Assume F has a mass point at $\hat{w}$: (Draw graph on whiteboard)

$$F(\hat{w}) = F(\hat{w}^-) + v_1(\hat{w}) \text{ where } v_1(\hat{w}) > 0$$

- Then average firm size l is discontinuous at $\hat{w}$: (Do on whiteboard)

$$l(\hat{w}^+) = l(\hat{w}) + v_2(\hat{w}) \text{ where } v_2(\hat{w}) > 0$$

- Then,

$$\begin{aligned}
\lim_{\epsilon\to 0} \pi(\hat{w}+\epsilon) - \pi(\hat{w}) &= \lim_{\epsilon\to 0}(y-\hat{w}-\epsilon)l(\hat{w}+\epsilon) - (y-\hat{w})l(\hat{w}) \\
&= \lim_{\epsilon\to 0}(y-\hat{w})(l(\hat{w}+\epsilon) - l(\hat{w})) - \epsilon l(\hat{w}+\epsilon) \\
&= \lim_{\epsilon\to 0}(y-\hat{w})(l(\hat{w}+\epsilon) - l(\hat{w})) \\
&= (y-\hat{w})v_2(\hat{w}) \\
&> 0
\end{aligned}$$

- A firm can make excess profits by offering wage $\hat{w}+\epsilon$ because
  - per-worker profit decreases continuously
  - firm competition decreases discretely

Step 3: $\underline{\mathbf{w}} = w_R$

- Step 1: $\mathbb{W}_F = [\underline{w}, \bar{w}] \subset [w_R, y]$
- Step 2: F is continuous:

$$\begin{aligned} l(w) &= \frac{k_u(1+k_e(1-F(W_R)))/(1+k_u(1-F(w_R)))}{(1+k_e(1-F(w)))(1+k_1(1-F(w^-)))} \\ &= \frac{k_u(1+k_e(1-F(w_R)))/(1+k_u(1-F(w_R)))}{(1+k_e(1-F(w)))^2} \end{aligned}$$

  for $w \geq w_R$.

- Moreover $F(\underline{w}) = 0$, which in turn implies $F(w_R) = 0$
- Hence,

$$l(\underline{w}) = \frac{k_u(1+k_e)/(1+k_u)}{(1+k_e)^2} = \frac{k_u/(1+k_u)}{(1+k_e)}$$

- Expected firm size at $\underline{w}$ is independent of the particular value of $\underline{w}$!
- Therefore, $\underline{w} = w_R$, since if $\underline{w} > w_R$, a firm could offer w_R and get the same amount of workers and higher per worker profits $\Rightarrow$ excess profits

Step 4: solve for F

- In equilibrium, $\pi(w) = \pi(w')$ for all $w, w' \in \mathbb{W}_F$
- In particular, $\pi(w) = \pi(w_R)$ for all $w \in \mathbb{W}_F$
- Solve $\pi(w) = \pi(w_R)$, using our solutions for $l(w), l(w_R)$:

$$\begin{aligned} (y-w)l(w) &= (y-w_R)l(w_R) \Leftrightarrow \\ (y-w)\frac{k_u(1+k_e)/(1+k_u)}{(1+k_e(1-F(w)))^2} &= (y-w_R)\frac{k_u/(1+k_u)}{(1+k_e)} \end{aligned}$$

- Do the algebra to find

$$F(w) = \frac{1+k_e}{k_e}\left[1-\left(\frac{y-w}{y-w_R}\right)^{1/2}\right] \tag{10}$$

- Using that $F(\bar{w}) = 1$, we see that $\bar{w} < y$:

$$\bar{w} = y - \frac{y-w_R}{(1+k_e)^2} < y \tag{11}$$

Step 5: solve for w_R

- Our reservation wage equation (4):

$$w_R - b = (\lambda_u - \lambda_e) \int_{w \geq w_R} \frac{1 - F(w)}{r + \sigma + \lambda_e(1 - F(w))} dw$$

- For simplicity, we use the assumption $r \to 0$ again:

$$w_R - b = (k_u - k_e) \int_{w \geq w_R} \frac{1 - F(w)}{1 + k_e(1 - F(w))} dw$$

- Plug in the solution of F:

$$w_R - b = \frac{(k_u - k_e)}{k_e} \int_{w_R}^{\bar{w}} \left[1 - \frac{1}{1 + k_e} \left(\frac{y - w}{y - w_R} \right)^{-\frac{1}{2}} \right] dw$$

- Solve the integral and use the solution to $\bar{w}$ to find:

$$w_R = \frac{(1 + k_e)^2 b + (k_u - k_e) k_e y}{(1 + k_e)^2 + (k_u - k_e) k_e} \tag{12}$$

- Summary: a continuous offer distribution with $b < \underline{w} = w_R < \bar{w} < y$

Model summary

- Model components: McCall + on-the-job search + fixed number of optimizing firms
- Importantly, all firms and households are ex ante identical
- Key result: a non-degenerate continuous wage distribution
  - Not possible in simple wage-posting model without on-the-job search
- Intuition?

Taking the model to the data

- Primitives: $b, \lambda_u, \lambda_e, \sigma, r$ - only one new parameter compared to McCall (but also one less!)
- λ_e can be identified from the frequency of job-to-job transitions
- An employed worker with wage w switches jobs at rate $\lambda_e(1 - F(w))$
- Total job switch rate: $\xi = \int_{w_R}^{\bar{w}} \lambda_e(1 - F(w))dG(w)$
- One can show that, in equilibrium, $\xi = \xi(\sigma, \lambda_e)$ (take-home exercise)
- λ_u is directly identified from the job-finding rate of unemployed workers, since unemployed workers accept all offers in this model

Predictions I: wage dispersion

- Wage dispersion: as in the McCall model, a formula for the mean-to-min ratio can be derived:

$$Mm_{BM} \approx \frac{\frac{\lambda_u-\lambda_e}{r+\sigma+\lambda_e}+1}{\frac{\lambda_u-\lambda_e}{r+\sigma+\lambda_e}+\rho} \qquad Mm_{McCall} = \frac{\frac{\lambda}{r+\sigma}+1}{\frac{\lambda}{r+\sigma}+\rho}$$

- $\approx$ comes from assuming $r \to 0$
- The BM model generates additional dispersion, as workers differ not only in their initial offers, but also in how far they have climbed up the job ladder
- Hornstein-Krusell-Violante (AER, 2011) report a monthly EE rate of 3.2 percent $\Rightarrow \lambda_e = 0.135$
- $Mm_{BM} = 1.22$, $Mm_{McCall} = 1.05$, $Mm_{data} > 1.8$
- Still not there, but a big step in the right direction
  - How does this relate to the results from the AKM regression?

Predictions II: wage correlations

- A worker with wage w finds a new job at rate:

$$\lambda_e^* = \lambda_e(1 - F(w)) \tag{13}$$

- Tenure is positively correlated with wage
  - From (13): higher $w \Rightarrow$ lower job-finding rate

- This challenges the interpretation of returns to tenure as reflecting productivity

- Firm size is positively correlated with wage
  - From (8): $I(w)$ is increasing in w, because workers continuously climb the job ladder

- Both are true in the data (Mortensen, Book 2003)

Taking stock

- Benchmark model for studying frictional wage dispersion, easily taken to the data
- New predictions: search frictions can rationalize
  - wage-tenure correlation
  - wage-firm size correlation
- We come closer to accounting quantitatively for residual wage dispersion
- BM also offers an interpretation of the AKM regression results
  - all wage dispersion stems from firm heterogeneity, even though all firms are identical in terms of productivity
  - firm fixed effects in AKM should not necessarily be interpreted as firm productivity differentials
- But, no role for search frictions in explaining
  - within-firm wage dispersion and, therefore, the individual fixed effects in AKM
  - downward wage mobility

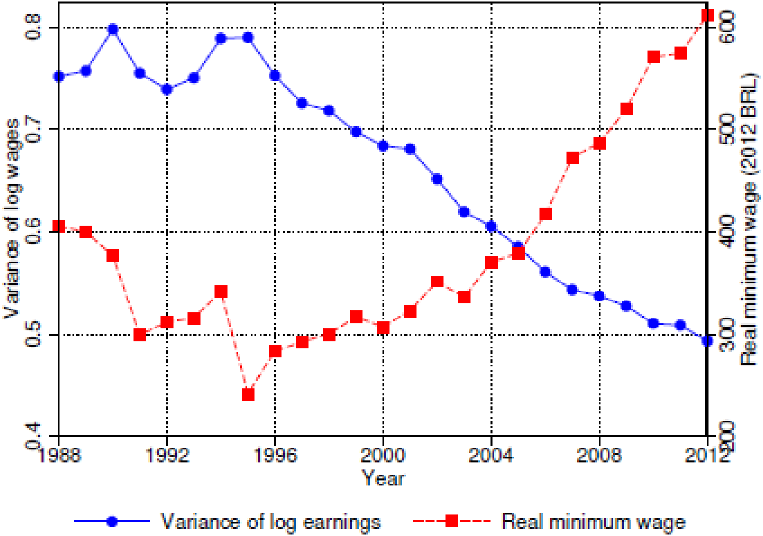
Extensions

- The basic BM model is too stylized to account for many empirical phenomena
- Because of its parsimony, the model can be extended in various directions: firm heterogeneity, worker heterogeneity, etc.
- For quantitative assessments, see Jolivet-Postel-Vinay-Robin (EER 2006), Mortensen (Book 2003), and Barlevy (ReStud 2008)

Recent applications

- Gottfries-Jarosch (2023) estimate a BM model for the US to quantify how much non-compete practices can suppress wages
  - A BM model with a finite number of firms and DRS production technology provides a natural laboratory for studying monopsony power
  - "Non-competes" are modeled as wage offers that do not allow job-to-job transitions
  - Banning "non-competes" raises average wages by 2-15% depending on labor market characteristics
- Engbom-Moser (AER 2023) estimate a BM model for Brazil, 1996-2012, following a reform that increased the mandated minimum wage:
  - Key fact: the entire wage distribution became more compressed after the reform, not just the lower end
  - BM theory: because firms set wage offers strategically in relation to $F(w)$, a higher minimum wage affects the entire distribution
  - They find that 70 percent of the drop in earnings inequality during this period can be attributed to the higher mandated minimum wage

Engbom-Moser (AER 2023)



Is wage posting reasonable benchmark?

- Key question: is wage posting a reasonable benchmark for the applications considered?
    - Bilateral bargaining?
        - See next lecture
    - Shouldn't contracts be allowed to be made contingent on observables?
        - Stevens (ReStud 2004) and Burdett-Coles (Econometrica, 2003) note that firms can reduce turnover and raise profits by offering wage-tenure contracts
    - Shouldn't firms be allowed to respond to employees' outside offers?
- Postel-Vinay and Robin (Econometrica, 2002) construct a job-ladder model extended with
    - firm and worker heterogeneity
    - allowing firms to make counteroffers
    - rationalizing within-firm wage dispersion among similar workers as well as downward wage mobility
    - offering a way to quantitatively separate the role of unobserved firm heterogeneity, unobserved worker heterogeneity, and luck in residual wage dispersion

Summary

- AKM: Reduced-form model for quantifying sources of residual wage dispersion
- Burdett-Mortensen: Benchmark model for studying frictional wage dispersion
- Key assumptions: on-the-job search + firm wage posting
- Key result: a non-degenerate distribution of wages across identical firms due to the trade-off in the wage-posting decision
  - Higher wage attracts more workers
  - Higher wage reduces per-worker profit
- We endogenized firm wage-setting behavior, but not the contact rate
- Next up: endogenizing the contact rate (and thinking about its implications for unemployment)